My chosen shape has just one adjustable handle: the width of the region that produces strong lensing. I call this the *lensing width*[29] and depict it in Figure 15.1.

How the Handles Influence the Wormhole's Appearance

Just as I had done for Gargantua (Chapter 8), I used Einstein's relativistic laws to deduce equations for the paths of light rays around and through the wormhole, and I worked out a procedure for manipulating my equations to compute the wormhole's gravitational lensing and thence what a camera sees when it orbits the wormhole or travels through it. After checking that my equations and procedure produced the kinds of images I expected, I sent them to Oliver and he wrote computer code capable of creating the quality IMAX images needed for the movie. Eugénie von Tunzelmann added background star fields and images of astronomical objects for the wormhole to lens, and then she, Oliver, and Paul began exploring the influence of my handles. Independently, I did my own explorations.

Eugénie kindly provided the pictures in Figures 15.2 and 15.4 for this book, in which we look at Saturn through the wormhole. (The resolutions of her pictures are far higher than my own crude computer code can produce.)

The Wormhole's Length

We first explored the influence of the wormhole's length, with modest lensing (small lensing width); see Figure 15.2.

When the wormhole is short (top picture), the camera sees one distorted image of Saturn through the wormhole, the primary image, filling the right half of the wormhole's crystal-ball-like mouth. There is